

BE 2nd Year CSE
Internal Test-1
Data Structure Using C
3CER3C3

Max Marks: 20

Time: 70 Mins

****Section A: Arrays (5 marks)** Attempt Any 2**

1. What is an array?
2. Explain the concept of array indexing.
3. What is a dynamic array? How does it differ from a static array?

****Section B: Searching (5 marks)** Attempt Any 2**

4. Compare linear search and binary search in terms of time complexity.
5. Describe binary search and its requirements.
6. How does binary search work on a sorted linked list?

****Section C: Sorting Techniques (5 marks)** Attempt Any 2**

7. Explain Merge sort algorithm.
8. Explain bubble sort with example and its algorithm.
9. Discuss the quick sort algorithm.

****Section D: Linked List (5 marks) Attempt Any 2**

10. Define a linked list. Explain how to create it.
11. How do you insert an element at the beginning of a linked list?
12. Explain the concept of a circular linked list.